3. Demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample.

CODING:

```
import math


data = [
    ['Sunny', 'Hot', 'No'],
    ['Sunny', 'Cool', 'Yes'],
    ['Rainy', 'Hot', 'No'],
    ['Rainy', 'Cool', 'Yes'],
    ['Overcast', 'Hot', 'Yes'],
    ['Overcast', 'Cool', 'Yes']
]


def entropy(rows):
    counts = {}
    for row in rows:
        counts[row[-1]] = counts.get(row[-1], 0) + 1

    result = 0
    for count in counts.values():
        p = count / len(rows)
        result -= p * math.log2(p)
    return result


def build_tree(rows, features):
    labels = [row[-1] for row in rows]
```

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```
if len(set(labels)) == 1:
    return labels[0]

if not features:
    return max(set(labels), key=labels.count)

best = max(
    range(len(features)),
    key=lambda i: entropy(rows) - sum(
        len([r for r in rows if r[i] == v]) / len(rows)
        * entropy([r for r in rows if r[i] == v])
        for v in set(r[i] for r in rows)
    )
)

feature = features[best]
tree = {feature: {}}

for value in set(row[best] for row in rows):
    subset = [r[:best] + r[best+1:] for r in rows if r[best] == value]
    tree[feature][value] = build_tree(
        subset, features[:best] + features[best+1:]
    )

  return tree

features = ['Weather', 'Temperature']
```

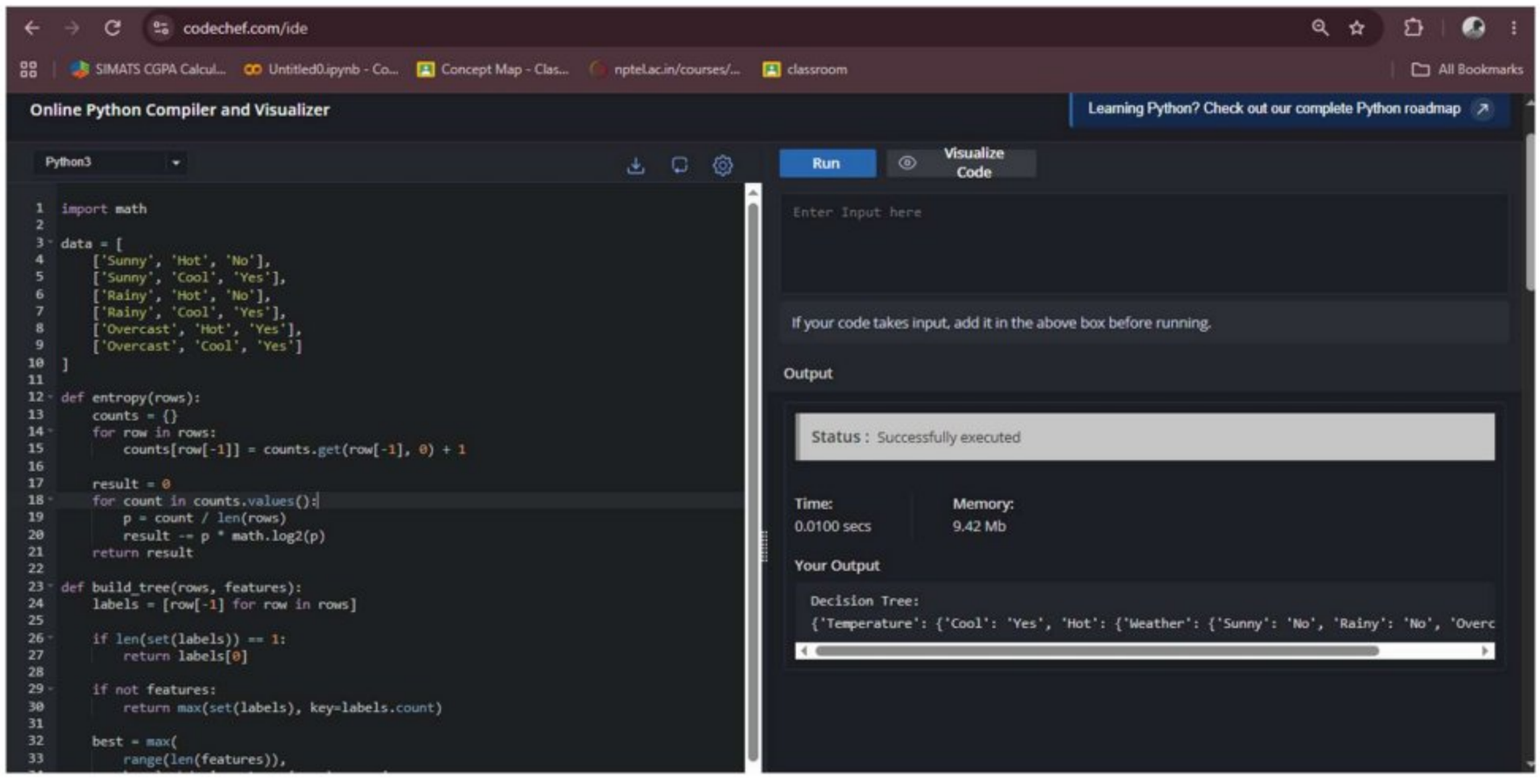
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```
tree = build_tree(data, features)


print("Decision Tree:")

print(tree)
```

OUTPUT:



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